

Name:	
Enrolment No:	

UPES

Mid Semester Examination, March 2025

Programme Name: B.Tech CSE	Semester: IV
Course Name : CSFS2003P	Time : 02 hrs
Course Code : Frontend Development	Max. Marks: 100
Nos. of page(s) : 3	
Calculator allowed: No	
Instructions: Please attempt according to the time provided and given weightage.	

SECTION A (30 Marks) 5 Questions Attempt all questions

S. No.		Marks	CO
Q 1	a) Define and explain CSS selectors. b) Differentiate between class selectors and id selectors.	6	CO1
Q 2	The CSS below is intended to style a button with rounded corners and a shadow effect but contains an error: <pre>.button { background-color: #4CAF50; color: white; padding: 10px 20px; borderradius: 5px; box-shadow: 0 2px 4px rgba(0,0,0,0.2); }</pre> a) Correct the error in the above code. b) Explain how the corrected property affects the button's appearance	2+4	CO1
Q 3	Discuss how external stylesheets improve project maintainability as compared to inline styles.	6	CO1
Q 4	a) Explain what responsive web design is. b) What role do CSS media queries play in creating responsive designs? Provide a brief description.	3+3	CO2
Q 5	a) Define HTML5 semantic markup and explain why it is important in modern web development. b) List three semantic elements introduced in HTML5 and briefly describe their typical usage	3+3	CO1

SECTION B (45 Marks) 3 questions

Q 6	a) Explain the importance of asset optimization (including images, fonts, etc.) and caching in web development. b) Describe important user-centric performance metrics .	7+8	CO2
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Q 7	Describe the fundamental CSS properties and methods available in Grid (e.g., grid-template-columns, grid-template-rows, grid-area, gap) versus those in Flexbox (e.g., flex-direction, justify-content, align-items, flex-wrap).	15	CO2
Q8	Using the below syntax, do the following a) Delete the paragraph with id = 'paragraph3' b) Hide the paragraph with id = 'paragraph1' <pre> index.html html> <head> <title>TestingExam goal</title> </head> <body> <h1>Test</h1> <div id = "primary"> <h2> Practical learning platform </h2> <p id = "paragraph1" style="color:red"> Ideal for beginners and intermediate learners </p> <p id = "paragraph2" style="color:blue"> Courses on basic programming, DSA, Analytics and Web development </p> <p id = "paragraph3" style="color:green"> A course for each semester </p> </div> <script src="index.js"></script> </body> </html> Add your code in a index.js file OR List and explain the critical components you would include (e.g., data visualizations, real-time statistics, user interaction elements) and why each is important. </pre>	15	CO2
SECTION C (25 Marks) 2 Questions Attempt any one question			
Q 9	Explain the key architectural differences among React, Angular, and Vue (e.g., component-based architecture, data binding strategies, and how each framework manages state and DOM updates).	25	CO3

OR

Below is a code snippet that includes an HTML form with an input field and a submit button. When the form is submitted, an arrow function should be executed that prevents the form's default behaviour and displays an alert with the text entered by the user. However, part of the arrow function is missing. Complete the blank (_____) in the JavaScript code to achieve the described functionality.

Username: Submit

Instructions:

- Replace the blank (_____) with the appropriate JavaScript code using an arrow function.
- Your solution should:
 - Select the input field by its id ("username").
 - Retrieve the value entered by the user.
 - Display an alert showing the retrieved value.
 - Additionally, explain briefly how your solution works.

```
<script>

// Attach an event listener to the form for the 'submit' event using an arrow function.
document.getElementById("myForm").addEventListener("submit", (event) => {

    // Prevent the default form submission behavior.
    event.preventDefault();

    // Fill in the blank: Use code to read the value from the input field with id "username"
    // and display it in an alert.

    _____;

});
</script>
```